

Amazon Sales Analytics & Customer Value Analysis

Executive Summary

This project analyzes 100,000 Amazon sales transactions to understand revenue performance, customer purchasing behavior, payment preferences, and customer value.

The analysis was conducted using Python, Pandas, NumPy, and Matplotlib to transform raw transaction data into actionable business insights.

Dataset Overview

Metric	Value
Total Orders	100,000
Total Customers	43,233
Total Revenue	\$91,825,647.92
Average Order Value	\$918.26
Analysis Period	2020–2024

Business Objectives

- Identify top-performing product categories and brands
 - Analyze revenue trends over time
 - Understand customer purchasing behavior
 - Evaluate customer lifetime value
 - Segment customers based on revenue contribution
 - Generate actionable business recommendations
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Revenue Analysis

Category Performance

Top revenue-generating categories:

- Electronics
- Sports & Outdoors
- Books
- Clothing
- Toys & Games
- Home & Kitchen

Revenue distribution across categories was relatively balanced, indicating a diversified product portfolio.

Brand Performance

Highest-performing brands included:

- CoreTech
- KiddoFun
- ReadMore
- UrbanStyle
- Zenith

These brands generated the largest share of total revenue and represent key growth drivers.

Customer Value Analysis

Customer Segmentation

Customers were segmented into four groups based on total spending:

Segment	Revenue Contribution
Bronze	\$4.94M
Silver	\$14.36M
Gold	\$25.48M
Platinum	\$47.05M

Key Finding

Platinum customers generated more than 50% of total revenue, highlighting the importance of retention and loyalty initiatives.

Payment Method Analysis

Revenue generated by payment method:

- Credit Card: \$32.1M
- Debit Card: \$18.5M
- UPI: \$13.9M
- Amazon Pay: \$13.7M
- Net Banking: \$9.1M
- Cash on Delivery: \$4.5M

Key Finding

Credit Cards remain the dominant payment method, while digital payment adoption continues to grow.

Geographic Analysis

Top-performing locations:

States

1. Texas (\$22.86M)
2. California (\$18.23M)

Cities

1. Charlotte
2. Dallas
3. San Jose
4. Seattle
5. Philadelphia

Key Finding

Revenue concentration in a small number of states suggests opportunities for targeted regional marketing.

Business Recommendations

Customer Retention

- Strengthen loyalty programs for Platinum customers.
- Develop targeted retention campaigns.

Revenue Growth

- Expand inventory for top-performing categories.
- Increase focus on high-performing brands.

Marketing Optimization

- Create personalized campaigns for Gold customers.
- Encourage progression from Gold to Platinum segments.

Payment Strategy

- Promote digital payment incentives.
- Reduce dependency on cash-on-delivery transactions.

Conclusion

This project demonstrates how data analytics can transform large-scale transactional data into actionable business insights. Through revenue analysis, customer segmentation, and visualization, key opportunities were identified to improve customer retention, optimize marketing efforts, and support revenue growth.